

ZHE CHEN

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EMPLOYMENT

- **Teaching and researching assistant** (A.T.E.R.) MAP5 (Université Paris Cité) September 2025- Current

EDUCATION

Ph.D., Applied Mathematics, October 2022- December 2025

Laboratoire Jacques-Louis Lions (Sorbonne Université),

Supervised by Frédérique Charles & François Golse

Master 2, Mathematics, Paris Dauphine University-PSL September 2021-September 2022

BS, Mathematics, Wuhan University September 2017-June 2021

TEACHING EXPERIENCE

- Supervision of undergraduate mathematical projects, Université Paris Cité, Bachelor 1 2025-2026
- Teaching assistant of **Ml02E015 Maths et Calcul 2** (60h), Université Paris Cité, Bachelor 1 2025-2026
- Teaching assistant of **Ml01E035 Maths et Calcul 1** (120h), Université Paris Cité, Bachelor 1 2025-2026
- Teaching assistant of **Complex Analysis FMA3Fo13EP**, École Polytechnique, Ingénieur 1A / Bachelor 3 2024-2025
- Teaching assistant of **Complex Analysis MAA313**, École Polytechnique, Ingénieur 1A / Bachelor 3 2023-2024

PUBLICATIONS AND PREPRINTS

- F. Charles, Z. Chen, and F. Golse, "Local Conservation Laws and Entropy Inequality for Kinetic Models with Delocalized Collision Integral" *Mathematical Models and Methods in Applied Sciences* (2026)
- Z. Chen, "A Formal Derivation of the Thick Spray Model from the Enskog-Boltzmann System" (Submitted)
- Z. Chen, "Conservative Formulations of the Standard Enskog and Povzner Equations" (Submitted)
- Doctoral thesis: "Study of Nonlocal Collisional Kinetic Models and Their Applications to Thick Sprays" (Defensed on 19 December 2025)

INTERNSHIP

- **Title:** The Grad limit for a system of soft sphere June 2022-September 2022
Supervisor: François Golse, Centre de mathématiques Laurent Schwartz, École Polytechnique
- **Title:** On a Two-Species Micro-Macro Model for Wormlike Micellar Solutions: Dynamic Stability Analysis April 2021 - June 2021
Supervisor: Ning Jiang, Wuhan University

CONFERENCES AND DOCTORAL SCHOOL ATTENDED

- IMT, Toulouse** MINT summer school & Particle Systems and PDEs XIV (*Selected Poster Presenter*) 15-26 June 2026
- HSM, Bonn** Special Topic School: Particles in flow. 23 -27 June 2025
- CIRM, Marseille** : Kinetic theory and fluid mechanics: couplings, scalings and asymptotics. 24-28 March 2025
- Aussois, France** Modeling, theory and numerics for PDEs: kinetic and hyperbolic systems 8-11 October 2024
- Chania, Greece** Summer school and research conference : Festum π 8-20 June 2024
- ETH Zürich** Kinetic and hydrodynamic PDEs : Conference in honor of François Golse's 60th birthday 16 - 19 January 2024
- ENS, Paris** Salve winter school 6-8 December 2023

Max-Planck institute, Leipzig Shocking Developments: New Directions in Compressible and Incompressible Flows: A Conference in Honor of Alexis Vasseur's 50th Birthday 26 - 30 June 2023

IHES, Bures-sur-Yvette Advances in Nonlinear Analysis and Nonlinear Waves, conference in honor of Frank Merle 22 - 26 May 2023

CIRM, Marseille Kinetic theory research school 14 - 18 November 2022

IST Austria 2nd IST Austria Summer School in Analysis and PDEs 25 - 29 July 2022

ETH Zürich When Kinetic Theory meets Fluid Mechanics 18 - 22 July 2022

HONORS

Recipient of SMAI BOUM support for a one-day workshop on kinetic theory, 2026

Sorbonne University doctoral contract funded by ED 386 October 2022-September 2025

The Paris Graduate School of Mathematical Sciences(PGSM) scholarship program 2021

The 1st Prize Scholarship of Mathematics Base Class 2020

TALKS AND SEMINAR

Summer School EUR MINT 2026 on Particle systems and PDEs, Institut de Mathématiques de Toulouse 15, June, 2026

Doctoral Working Group in Harmonic Analysis and Numerical Analysis & PDEs, LMO, Université Paris Saclay 5, May, 2026

PhD Defense, Sorbonne Université 19, December, 2025

Working seminar Analyse, Modelisation & Simulation, MAP5, Université Paris Cité 19, September, 2025

Working seminar for Ph.D. students, LJLL 19, March, 2025

Second year PhD student presentation, CMLS 11, June, 2024

Organizer of First-year PhD student research talks, LJLL 23, November, 2023

Organizer of Working seminar for Ph.D. students in the LJLL (Groupe de Travail des Thésard-es du LJLL) May, 2023-July, 2024

Host of Advanced Analysis Reading Group, Wuhan University September, 2018-April, 2019

Host of Mathematical Analysis and Linear Algebra Reading Group, Wuhan University September, 2017-April, 2018

SKILLS

Software MATLAB, $\text{\LaTeX}$.

Languages Mandarin(Native), English(Academic), French(Intermediate)